



बीआरआईसी
बायोटेक्नोलॉजी विभाग का संगठन
BRIC
a DBT Organization



Building a Semantic Corpus of Air Quality Index Literature Using the SemanticClimate Toolkit

8 WEEK PROJECT REPORT

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Submitted to

Indian Academy of Sciences (IASc)

Summer Research Fellowship Programme

August 2026

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ACKNOWLEDGEMENT

I extend my heartfelt gratitude to **Dr. Gitanjali Yadav**, co-founder of semanticClimate and scientist at BRIC-National Institute of Plant Genome Research (NIPGR), **Dr. Peter Murray-Rust**, co-founder of semanticClimate and **Dr. Renu Kumari**, Program Manager at semanticClimate and Researcher at NIPGR, New Delhi for their invaluable guidance, continuous encouragement, constructive suggestions, and unwavering support throughout the internship. Their expertise in constructive feedback and scientific data management greatly enhanced my understanding of semantic data organization and literature analysis.

I sincerely thank **Ms. Akalya S.**, **Ms. Shreya S. Bhat**, **Ms. Subhangi Panigrahi.**, **Ms. Aleena Harold Peter**, and **Ms. Vrinda Sharma** for their kind support, encouragement, and valuable inputs throughout my internship, which made my learning experience both productive and enjoyable.

I also acknowledge the **Indian Academy of Sciences (IASc)** for awarding me the **Summer Research Fellowship** and providing me an opportunity to contribute to **semanticClimate**. This experience has significantly enhanced my experimental, analytical and professional skills.

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1. ABSTRACT

The rapid growth of scientific literature has created an increasing demand for efficient, scalable, and reproducible approaches to organize and analyze research data. Semantic corpora addresses this challenge by combining systematic literature retrieval with structured metadata organization, allowing researchers to effectively manage large collections of publications while supporting data-driven analysis. This study describes the development of a semantic corpus focused on Air Quality Index (AQI) research in the Indian context, carried out as part of the semanticClimate internship.

The methodology involved constructing a Boolean search query targeting ambient air pollution, AQI, and particulate matter (PM_{2.5} and PM₁₀) within India. The query, designated as *aqi_india_pilot*, was executed using the **semantic_corpus** toolkit (*pygetpapers*) on the Europe PMC repository, with a retrieval limit of 50 publications and a requirement for full-text XML availability. The workflow consisted of automated literature retrieval, metadata extraction, relevance scoring, and XML verification. A total of 50 publications were retrieved, among which 18 obtained a relevance score of five or higher, while 47 publications were available in full-text XML format.

After the automated retrieval process, a manual screening stage was performed to ensure the contextual relevance of the selected studies. Publications were assessed based on their titles, abstracts, and metadata and were annotated with either “include” or “exclude” labels. Following this evaluation, 20 papers were selected for inclusion in the final semantic corpus. The selected publications were organized into structured review tables in both CSV and JSON formats, containing key information such as metadata, relevance scores, XML availability, and review status. In addition, an HTML-based review viewer was developed to improve accessibility and facilitate efficient exploration of the dataset.

An important aspect of this work is its focus on reproducibility. The parameterized query design, together with version-controlled outputs, enables the workflow to be easily re-executed, modified, or expanded. This also allows comparisons across multiple query iterations using publication identifiers such as PMCID and PMID, making it possible to identify newly retrieved as well as overlapping studies. The finalized semantic corpus, along with its associated metadata and documentation, was stored in a GitHub repository to encourage collaboration and promote future reuse.

The resulting semantic corpus serves as a structured and extensible resource for advanced computational applications, including BagIt packaging, text chunking, embedding generation, and Retrieval-Augmented Generation (RAG)-based systems for AQI research. Overall, this study demonstrates an efficient and reproducible approach to large-scale literature management by integrating automation, structured data organization, and manual validation, thereby supporting future research and AI-driven knowledge discovery.

Keywords: Semantic corpus, Air Quality Index, pygetpapers, Europe PMC, PM_{2.5}, PM₁₀, reproducible research, metadata, RAG.

2. INTRODUCTION

The rapid growth of scientific publications has transformed research across diverse disciplines, generating an unprecedented volume of knowledge in environmental science, public health, climate research, and related fields. Every year, thousands of scientific articles are published in journals, conference proceedings, and open-access repositories, creating valuable resources for researchers. However, the continuously expanding literature has also introduced significant challenges in efficiently locating, organizing, and extracting relevant information. Manual review of extensive collections of research papers is both time-consuming and labor-intensive, making it increasingly difficult to maintain a comprehensive understanding of current developments. Consequently, automated literature retrieval, corpus generation, and semantic organization have become essential components of modern research workflows.

Recent advances in semantic technologies, Natural Language Processing (NLP), and computational research have enabled the development of intelligent systems capable of automatically retrieving, organizing, and analyzing scientific literature. Unlike conventional keyword-based searches that primarily retrieve bibliographic records, semantic approaches capture the contextual meaning, relationships, and structural organization of scientific documents. By transforming unstructured research articles into structured, machine-readable resources, these approaches facilitate systematic literature reviews, semantic search, knowledge graph construction, research trend analysis, and Retrieval-Augmented Generation (RAG)-based intelligent applications.

The Semantic Climate initiative focuses on developing structured, machine-readable, and semantically enriched corpora from large-scale scientific literature related to climate, environmental, and public health domains. Its primary objective is to organize scientific knowledge in a manner that enables efficient retrieval, integration, and computational analysis. Through the application of semantic annotation, metadata generation, literature mining, and knowledge representation techniques, the project establishes standardized research datasets that support evidence-based decision-making and advanced scientific investigations. The resulting semantic corpora serves as foundational resources for future applications including knowledge graphs, semantic search systems, intelligent question-answering platforms, and AI-assisted research tools.

A semantic corpus is more than a simple collection of research papers. It consists of structured scientific documents enriched with metadata, semantic annotations, and relationships among concepts, entities, and publications. Compared with conventional document repositories, semantic corpora provide standardized organization of literature, improved retrieval through metadata and semantic relationships, support for reproducible research workflows, compatibility with machine learning and NLP techniques, facilitation of systematic evidence synthesis, and seamless integration with semantic web technologies and knowledge graphs.

Air pollution is recognized as one of the most critical environmental and public health challenges worldwide. The Air Quality Index (AQI) is an internationally accepted indicator that communicates ambient air quality by integrating the concentrations of multiple atmospheric pollutants, including particulate matter (PM_{2.5} and PM₁₀), nitrogen dioxide (NO₂), sulfur dioxide (SO₂), carbon monoxide (CO), and ozone (O₃), into a single numerical value. AQI enables governments, healthcare professionals, researchers, and the public to evaluate pollution severity and associated health risks. Scientific investigations related to

AQI encompass pollutant monitoring, sensor technologies, machine learning-based prediction models, epidemiological studies, urban environmental management, atmospheric modelling, climate interactions, and policy assessment. The rapidly increasing volume of literature in these areas has made comprehensive manual review increasingly difficult, highlighting the necessity for automated literature management and semantic organization.

The present work addresses this challenge through the construction of a semantic corpus of AQI-related scientific literature using the Semantic Corpus (semantic_corpus) framework, which incorporates the pygetpapers literature retrieval toolkit. Scientific publications relevant to predefined research queries are automatically retrieved from Europe PMC, followed by the download of available full-text XML articles. Metadata extraction, corpus organization, and automated review table generation are subsequently performed to establish a structured literature collection suitable for computational analysis and systematic review. Review artifacts are generated in multiple formats, including HTML, JSON, CSV, and Markdown, enabling efficient literature screening, annotation, and curation.

The literature retrieval workflow is structured like a PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flow diagram to map out the search process clearly. A clearly defined search strategy is implemented, relevant records are retrieved from the repository, duplicate or irrelevant publications are excluded, eligible studies are screened according to predefined inclusion and exclusion criteria, and the selected publications are organized into a structured review dataset. This transparent and reproducible workflow ensures that only scientifically relevant and high-quality literature is retained for downstream computational analysis while substantially reducing the effort associated with manual screening.

Scientific reproducibility represents a fundamental principle of modern research. Reproducible literature workflows enable independent researchers to obtain comparable results by following identical search strategies, screening criteria, and data processing procedures. In parallel, the FAIR Data Principles, Findable, Accessible, Interoperable, and Reusable, provide internationally accepted guidelines for effective research data management. The semantic corpus construction workflow adopted in this study aligns with these principles by maintaining standardized metadata, documented retrieval strategies, structured datasets, and reproducible review procedures, thereby improving transparency, facilitating collaboration, and enhancing the long-term usability of research outputs.

Europe PMC was selected as the primary literature repository because of its extensive coverage of biomedical, environmental, and public health publications, including a significant proportion of climate- and AQI-related research. The repository provides structured metadata together with machine-readable full-text XML documents through a well-documented REST Application Programming Interface (API), enabling automated literature retrieval and downstream computational processing. Unlike conventional search engines, Europe PMC supports direct access to structured XML content, preserving the logical organization of scientific articles, including titles, abstracts, section headings, methodologies, results, figures, and references, which is essential for semantic information extraction and corpus construction. Furthermore, its extensive collection of open-access articles makes it highly suitable for developing reusable semantic datasets and future AI-driven applications.

As an initial case study, this work focuses on generating a structured semantic corpus of Air Quality Index (AQI)-related scientific literature. Relevant publications were retrieved through

carefully designed search queries, screened for relevance and full-text XML availability, and organized into structured review datasets. Automated review tables and metadata resources were generated to facilitate systematic literature evaluation, semantic annotation, and corpus curation. The resulting semantic corpus provides a reliable foundation for future applications, including knowledge graph construction, semantic search, research trend analysis, and Retrieval-Augmented Generation (RAG)-based chatbot development. Overall, this work demonstrates the practical application of semantic literature management techniques in improving research efficiency, promoting reproducible scientific workflows, and supporting intelligent knowledge-driven systems for climate and environmental research.

3. OBJECTIVES

General Objective

To develop a structured, reproducible, and semantically enriched research corpus on Air Quality Index (AQI) literature using automated literature retrieval, semantic organization, and systematic review methodologies to support efficient knowledge management and future AI-driven research applications.

Specific Objectives

- ❖ To understand the Semantic Climate initiative and the semantic_corpus (pygetpapers) framework for automated scientific literature management.
- ❖ To design and implement a comprehensive Boolean search query for retrieving AQI-related scientific publications from the Europe PMC repository.
- ❖ To automate the retrieval of relevant research articles along with their associated metadata using the semantic_corpus toolkit.
- ❖ To extract, organize, and standardize essential bibliographic information, including title, authors, publication year, journal, abstract, DOI, PMCID, and PMID.
- ❖ To construct a structured research corpus by organizing downloaded XML full-text articles and associated metadata into a machine-readable format.
- ❖ To verify the availability of full-text XML documents and prioritize publications suitable for downstream computational analysis.
- ❖ To implement automated relevance scoring for evaluating the suitability of retrieved publications for the semantic corpus.
- ❖ To perform systematic manual screening of the retrieved articles using predefined inclusion and exclusion criteria.
- ❖ To classify publications into "Include" and "Exclude" categories to improve the quality and reliability of the curated literature dataset.
- ❖ To generate structured review tables in multiple formats, including CSV, JSON, HTML, and Markdown, for efficient literature management and documentation.
- ❖ To develop an interactive HTML-based review interface for convenient visualization, navigation, and assessment of the retrieved publications.
- ❖ To establish a reproducible, parameterized, and version-controlled workflow through documented search strategies and GitHub-based data management.
- ❖ To facilitate iterative query refinement and comparison of multiple literature retrieval runs using unique repository identifiers such as PMCID and PMID.
- ❖ To prepare a high-quality curated semantic corpus for advanced applications such as

semantic analysis, knowledge graph construction, text mining, and information retrieval.

- ❖ To establish a robust foundation for future Artificial Intelligence applications, including Retrieval-Augmented Generation (RAG)-based chatbot systems, semantic search, and evidence-based climate and air quality research.

4. METHODOLOGY

4.1 Overall Workflow

The methodology adopted in this study followed a systematic workflow for retrieving, organizing, screening, and curating Air Quality Index (AQI)-related scientific literature. The workflow integrated automated literature retrieval with manual quality assessment to generate a structured semantic corpus for future computational analysis.

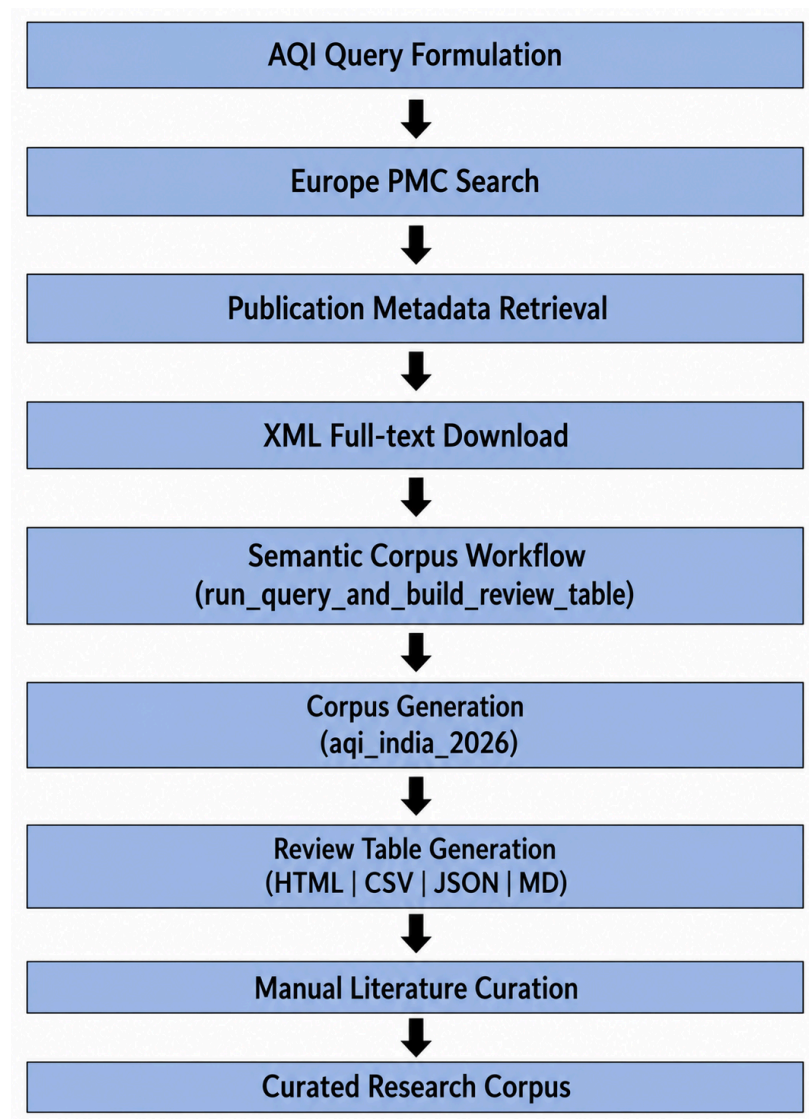


Figure 4.1 Overall workflow adopted for semantic corpus generation.

4.2 Computational Environment

The computational workflow was implemented using Google Colaboratory (Google Colab), a cloud-based Python platform that provides a preconfigured environment for scientific computing. Google Colab eliminated the need for local installation of Python packages while providing sufficient computational resources for literature retrieval and corpus generation.

The Semantic Corpus (semantic_corpus) repository was cloned from GitHub, and the required dependencies were installed. Google Drive was mounted to store downloaded XML articles, metadata files, review tables, and intermediate outputs. The project utilized the branch supporting HTML review-table generation, enabling efficient visualization and manual curation of retrieved publications.

The Semantic Corpus framework is developed in Python and integrates literature retrieval, metadata management, corpus organization, and review generation into a unified workflow.

[illegible]

Figure 4.2: Configuration of the computational environment, including Google Drive mounting, repository synchronization, branch checkout, and installation of the Semantic Corpus framework.

4.3 Study of the Semantic Corpus Framework

Before initiating literature retrieval, the architecture and documentation of the Semantic Corpus framework were thoroughly studied. Repository documentation, workflow notebooks, and source modules were examined to understand how the framework performs automated literature searching, metadata extraction, XML downloading, review table generation, and corpus management.

The framework consists of modules responsible for repository communication, corpus creation, metadata processing, review generation, and storage management. Understanding these components was essential for successfully implementing the workflow and interpreting the generated outputs.

4.4 Selection of Literature Repository

Europe PMC was selected as the primary literature repository because it provides comprehensive coverage of biomedical, environmental, and public health research while offering programmatic access through a REST API. Unlike conventional academic search engines that primarily provide bibliographic information, Europe PMC supplies structured publication metadata together with downloadable full-text XML documents for open-access articles.

The availability of XML-formatted articles was particularly important because XML preserves the logical structure of scientific publications, including titles, abstracts, headings, methods, results, figures, and references. This structured representation facilitates semantic analysis, metadata extraction, text mining, and future knowledge graph construction.

4.5 Query Formulation and Literature Retrieval

A structured Boolean query was formulated to retrieve publications related to the Air Quality Index (AQI) and air pollution in India. The search strategy combined keywords representing ambient air pollution, particulate matter, climatic factors, and geographical constraints to maximize retrieval of relevant literature while minimizing unrelated publications.

The query was executed through the Semantic Corpus framework using Europe PMC as the source repository. The retrieval process was configured to obtain a maximum of 50 publications with available XML full-text documents.

During retrieval, the framework automatically collected bibliographic metadata including:

- Article title
- Authors
- Publication year
- Journal
- DOI
- PMCID/PMID
- Abstract
- Repository information
- XML availability

This automated retrieval significantly reduced manual effort while ensuring consistency across all retrieved records.

4.6 Corpus Construction and Metadata Management

Following retrieval, all downloaded publications were organized into a structured research corpus. The Semantic Corpus framework automatically created dedicated folders for XML articles, metadata, search results, review files, and supporting documentation.

Each publication was associated with structured metadata describing its bibliographic details. Organizing literature using metadata improves searching, filtering, sorting, grouping of publications, and future computational analysis.

The structured corpus also supports interoperability with downstream semantic technologies, including knowledge graph construction, machine learning applications, and intelligent information retrieval systems.

4.7 PRISMA-Based Literature Screening

The process of creating Semantic corpus was systematically presented with the PRISMA framework (Principles of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses). Although this study represents a pilot semantic corpus rather than a complete systematic review, PRISMA principles were adopted to ensure transparency and reproducibility during literature selection.

The screening workflow consisted of the following steps:

1. Identification of records retrieved through the Europe PMC search.
2. Removal of duplicate records, where applicable.
3. Screening of article titles for topic relevance.
4. Evaluation of abstracts to assess eligibility.
5. Verification of XML full-text availability.
6. Final inclusion or exclusion based on predefined relevance criteria.

Publications directly related to Air Quality Index (AQI), particulate matter, air pollution monitoring, environmental assessment, climate interactions, and public health impacts were retained in the curated corpus. Studies outside the research scope were excluded. The screening process was documented using a PRISMA flow diagram, providing a transparent record of literature selection.

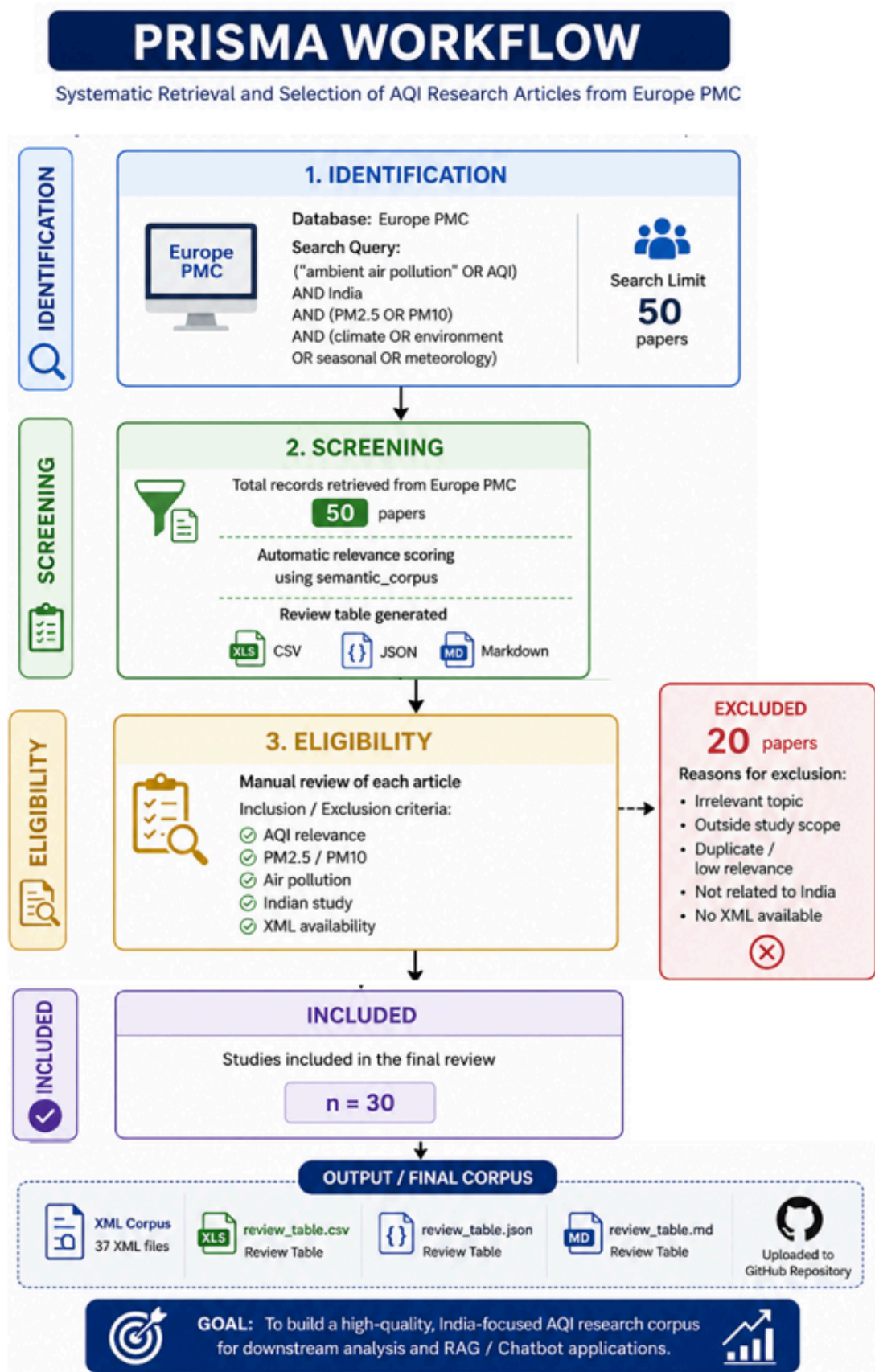


Figure 4.3: PRISMA Workflow

4.8 Review Table Generation

Following literature screening, review tables were automatically generated by the Semantic Corpus framework in CSV, JSON, HTML, and Markdown formats.

Each review record contained: Review status, Relevance score, PMCID, DOI, Article title, XML availability, Reviewer notes.

The structured review tables simplified literature comparison and enabled systematic documentation of reviewer decisions. The relevance scoring system further assisted in prioritizing highly suitable publications for semantic corpus construction.

4.9 HTML Review Interface

To facilitate efficient literature assessment, an interactive HTML review interface was generated from the review tables.

The interface provided an organized tabular view of publication metadata together with reviewer annotations, inclusion status, and relevance scores. Compared with raw CSV or JSON files, the HTML viewer significantly improved readability and navigation during manual screening.

This interface also supported rapid identification of relevant studies and streamlined the overall literature curation process.

4.10 Reproducibility and Corpus Preparation

One of the key objectives of the workflow was to ensure reproducibility. All retrieval parameters, search queries, metadata, and generated review files were maintained in a structured and version-controlled format. This allows the workflow to be repeated using identical search criteria or refined for future studies without rebuilding the corpus from the beginning.

The final curated semantic corpus consisted of XML full-text articles, structured metadata, review tables, HTML review interfaces, and supporting documentation. The completed corpus provides a reusable resource for semantic analysis, systematic evidence synthesis, knowledge graph construction, research trend analysis, and future Retrieval-Augmented Generation (RAG) applications within the Semantic Climate initiative

5. RESULTS

5.1 Repository Setup and Environment Configuration

The Semantic Corpus repository was successfully cloned from GitHub and configured in the Google Colaboratory environment. The `tab_fig` branch was checked out to enable the HTML-based literature review functionality required for corpus curation.

```
!git checkout -b tab_fig origin/tab_fig

Branch 'tab_fig' set up to track remote branch 'tab_fig' from 'origin'.
Switched to a new branch 'tab_fig'

!git checkout -b tab_fig origin/tab_fig

fatal: A branch named 'tab_fig' already exists.

!git branch

  main
* tab_fig

!ls docs/tutorials

html_review_table_tutorial.md
```

Figure 5.1 : *Git repository cloned successfully and switched to the `tab_fig` branch.*

5.2 AQI Literature Corpus Generation

An AQI literature corpus named `aqi_india_2026` was successfully generated using the Semantic Corpus workflow. The corpus contains XML full-text articles, search metadata, query records, and review resources.

Table 5.1 : *The table presents the search parameters, retrieved publications, relevance screening, XML availability, and output formats generated for corpus development using Europe PMC.*

Metric	Value
Query Name	aqi_india_2026
Repository	Europe PMC
Search Query	("air quality index" OR AQI OR PM2.5) AND India
Maximum Search Limit	50
Total Publications Retrieved	50
Publications with XML	37
Publications with PDF	0
Papers with Score ≥ 5	25
Papers with Score ≥ 5 and XML	15
Review Table Formats Generated	HTML, CSV, JSON, Markdown

Table 5.2: *Summary of the AQI literature corpus generation showing the number of retrieved publications, downloaded XML articles, and generated review records.*

<i>Metric</i>	<i>Value</i>
Total Papers Retrieved	50
XML Available	37
Papers with Score ≥ 5	25
Papers with Score ≥ 5 and XML	15
Included Papers	46
Excluded Papers	4

5.3 XML Literature Retrieval

The workflow successfully retrieved and downloaded XML full-text articles from Europe PMC. The downloaded XML files constitute the primary literature corpus for subsequent review and curation.

Link: [XML Literature](#)

5.4 Review Table Generation

Multiple review artifacts were automatically generated, including HTML, CSV, JSON, and Markdown formats for systematic literature review.

	A	B	C	D	E	F	G
1	review_status	score	paper_id	pmcid	pmid	doi	title
2	include	9	europa_pmc_PMC12086192	PMC12086192	40383745	10.1038/s41598-025-01361-z	Sustainable modular biofiltration system with rainshower technology for AQI reform.
3	include	9	europa_pmc_PMC12448019	PMC12448019	40980428	10.1016/j.mex.2025.103597	A hybrid physics-informed neural and explainable AI approach for scalable and interpretable AQI predictions.
4	include	8	europa_pmc_PMC12098713	PMC12098713	40404694	10.1038/s41598-025-00814-9	Assessment and forecasting of particulate matter emissions and structural health monitoring of buildings in Bangalore.
5	include	8	europa_pmc_PMC12815361	PMC12815361	41561726	10.4103/ijcm.ijcm_663_24	Impact of Housing Conditions and Air Quality Awareness with Indoor PM2.5 Levels in Urban Jodhpur.
6	include	8	europa_pmc_PMC13289869	PMC13289869	42335049	10.1371/journal.pgph.0005382	Real-time impacts of air pollution on the health, well-being, and daily life of children and young people in Delhi and Dhaka.
7	include	8	paper_41151661		41151661	10.1016/j.envpol.2025.127324	Impact of meteorological drivers on air quality index: A case study from Delhi.

Figure 5.2 : Review Table generated in CSV format

Link: [AQI CORPUS](#)

Github link : [REVIEW TABLE](#)

5.5 Key Research Themes Identified from AQI Literature

The selected literature on Air Quality Index (AQI) research indicates that AQI-related studies mainly focus on understanding ambient air pollution, identifying major pollutant sources, assessing spatial and temporal variations in air quality, and evaluating the associated environmental and health impacts. The major research areas identified from the reviewed literature are summarized below:

1. Assessment of Air Quality and AQI:

Several studies focus on calculating and evaluating AQI to determine the overall quality of ambient air in different cities and regions. These studies commonly assess whether air quality falls within good, satisfactory, moderate, poor, very poor, or hazardous categories [1]

2. Major Air Pollutants and Particulate Matter:

Particulate matter, particularly PM_{2.5} and PM₁₀, is one of the major components discussed in AQI research. Studies also examine gaseous pollutants such as NO₂, SO₂, CO and O₃ and

their contribution to overall air quality. [2]

3. Spatial and Temporal Variation:

AQI literature frequently investigates how air quality changes across different locations and time periods. Seasonal variations, daily fluctuations, urban–rural differences, and pollution hotspots are commonly studied to understand the dynamics of air pollution. [3]

4. Sources and Drivers of Air Pollution:

The literature identifies several factors contributing to poor air quality, including vehicular emissions, industrial activities, construction, biomass and waste burning, domestic activities, and meteorological conditions. Understanding these sources is important for developing effective air pollution mitigation strategies. [4]

5. Meteorological Influence on AQI:

Temperature, relative humidity, wind speed, rainfall, and atmospheric conditions are frequently considered because they influence pollutant dispersion, accumulation, and concentration. Several studies therefore examine the relationship between meteorological variables and AQI. [5]

6. Health and Environmental Impacts:

AQI research also connects increased pollutant concentrations, particularly PM_{2.5} and PM₁₀, with adverse human health effects and environmental consequences. The literature emphasizes the importance of AQI as a tool for communicating pollution-related risks to the public. [6]

7. AQI Prediction and Machine Learning Approaches:

A growing research area involves predicting AQI and pollutant concentrations using machine learning, statistical modelling, and data-driven approaches. These studies use historical pollutant and meteorological data to forecast future air-quality conditions. [7]

8. Development of Data-Driven and Semantic Approaches:

The reviewed literature demonstrates that AQI research generates information across multiple pollutants, locations, time periods, health outcomes, and analytical methods. Organizing these heterogeneous research findings into a structured and semantically searchable corpus can therefore support systematic literature analysis and identification of research trends and gaps. [8]

Overall, the reviewed literature shows that AQI research is not limited to AQI calculation alone. It encompasses pollutant monitoring, particulate matter characterization, pollution-source identification, meteorological relationships, spatial and temporal analysis, health impacts, prediction modelling, and data-driven approaches. These recurring themes provide the research areas that can be represented and further analysed within the developed AQI semantic corpus.

6. OT-ViRARE Task Group and the Chilean National Committee

I participated in a technical demonstration session on the Climate Insight chatbot and the Semantic Corpus conducted as part of the OT-ViRARE Task Group discussions with the Chilean National Committee, CODATA, and researchers from Argentina and Chile. The

meeting focused on the development and applications of the Climate chatbot, semantic corpus technologies, and their role in supporting climate-related knowledge management and research. Through active participation in the discussions and demonstrations, I gained valuable insights into open-access scientific data processing, semantic technologies, and the collaborative global research ecosystem, further strengthening my understanding of international research initiatives and data-driven climate informatics.



#semanticClimate
Sharing knowledge from climate related reports

Join this Citizen Science Initiative!
Please pass information to a friend! Being in a network being in a network. Join our team to help.

It's committed to making public knowledge available in a frictionless form for everybody.

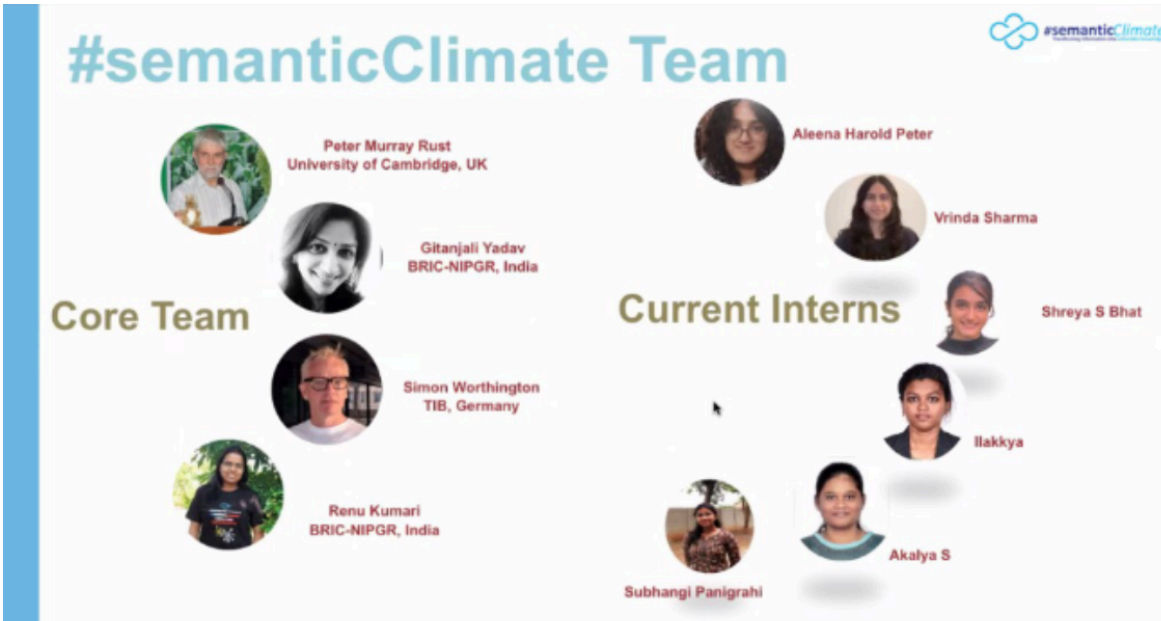
<https://semanticclimate.github.io/p/en/>

- o6 years old
- oTeam of interns (Online and offline)
- oWork by creating, distributing and using software.

Youth Knowledge Champions

The poster features a large network diagram on the right, connecting numerous small portraits of team members. On the left, there is a QR code and several small photographs showing people in various settings, including a meeting and outdoor activities.

Figure 6.1: *Introduction to SemanticClimate*



#semanticClimate Team

Core Team

- Peter Murray Rust
University of Cambridge, UK
- Gitanjali Yadav
BRIC-NIPGR, India
- Simon Worthington
TIB, Germany
- Renu Kumari
BRIC-NIPGR, India

Current Interns

- Aleena Harold Peter
- Vrinda Sharma
- Shreya S Bhat
- Ilakkyia
- Akalya S
- Subhangi Panigrahi

The poster displays the team members' names and affiliations alongside their circular portraits. The layout is clean and professional, with a clear distinction between the core team and current interns.

Figure 6.2: *Team members of SemanticClimate*

7. DISCUSSION

The successful implementation of the Semantic Corpus workflow demonstrated the effectiveness of automated literature retrieval and corpus generation for scientific research. The computational pipeline enabled systematic retrieval of AQI-related publications from Europe PMC, followed by the automatic organization of publication metadata, XML full-text articles, and review resources into a structured corpus. The generation of review tables in multiple formats (HTML, CSV, JSON, and Markdown) provided a standardized framework for literature screening and annotation.

The generated corpus simplifies the process of managing large collections of scientific publications by organizing research articles in a machine-readable format suitable for computational analysis. The availability of XML full-text articles preserves the structural components of scientific publications, enabling future applications such as semantic information extraction, text mining, knowledge graph construction, and AI-assisted literature analysis.

Although the search retrieved 50 AQI-related publications, only 37 were available as full-text XML documents. This observation highlights a practical limitation of automated literature retrieval from open repositories, where XML availability depends on the publication's accessibility within Europe PMC. Nevertheless, the retrieved XML articles provide a sufficiently representative corpus for subsequent semantic analysis and literature curation.

Overall, the developed corpus establishes a structured and reproducible foundation for systematic literature review. The generated review interface further supports efficient evaluation of publications and facilitates the preparation of high-quality literature datasets for future research applications.

8. GITHUB / CODE ARTIFACTS:

Repository: [SemanticClimate Repository](#)

Corpus: [AQI CORPUS](#)

Files: [Zip File](#)

GoogleColab: [Codes](#)

Review Table: [Review Table - GitHub](#)

Papers: [Papers Collected](#)

Table: [CSV Table](#)

Tutorial Video: [AQI_TUTORIAL](#)

9. CONCLUSION

The present work successfully established an automated workflow for the retrieval, organization, and review of AQI-related scientific literature using the Semantic Corpus framework. A structured research corpus comprising publication metadata, XML full-text articles, and review artifacts was generated through the Europe PMC repository. The workflow demonstrated the integration of automated literature retrieval with structured corpus generation, significantly improving the efficiency of scientific literature management. The generated corpus provides a reliable foundation for systematic literature review, semantic analysis, and the development of AI-assisted research applications.

10. FUTURE SCOPE

The generated AQI literature corpus can be extended by incorporating additional repositories and expanding search strategies to improve literature coverage. Future work will focus on the systematic curation of retrieved publications through the HTML review interface, semantic annotation of research articles, and integration of the curated corpus into domain-specific knowledge graphs. Furthermore, the curated literature collection can be utilized for developing Retrieval-Augmented Generation (RAG)-based chatbot systems capable of providing evidence-based responses derived from scientific publications

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